Transformer Architectures: A Comprehensive Study

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Abstract

We present a novel approach to transformer architectures that achieves state-of-the-art results on standard benchmarks. Our method introduces a new training objective that improves convergence by 44% while reducing computational cost by 31%. Experiments on 3 datasets confirm consistent improvements over strong baselines. Code will be released at github.com/bfai-research/transformer-architectures.

1. Introduction

The field of transformer architectures has seen rapid progress in recent years. However, key challenges remain around scalability, generalization, and computational efficiency. This work addresses these gaps through a principled framework that builds on prior work while introducing key innovations in the training procedure and model architecture.

2. Related Work

Seminal work by Vaswani et al. (2017) laid the foundation for modern transformer architectures research. Subsequent contributions from Brown et al. (2020), Wei et al. (2022), and Ouyang et al. (2022) demonstrated the power of scale and instruction tuning. Our work differentiates by focusing on sample efficiency and robustness, areas less explored in the prior literature.

3. Methodology

We propose a 24-layer architecture with 758 hidden dimensions. Training uses a batch size of 1024 tokens with a cosine learning rate schedule over 140k steps. The key innovation is a modified attention kernel that reduces the quadratic complexity to $O(n \log n)$.

4. Experiments

We evaluate on 3 benchmarks achieving 80.1% accuracy on the primary task. Ablations confirm each component contributes positively. Inference throughput is 5x faster than the baseline.

5. Conclusion

We presented a new approach to transformer architectures that advances the state of the art while improving efficiency. Future work will explore scaling to larger models and extending to multilingual and multimodal settings.